

Satvik Jambagi

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EDUCATION

Vellore Institute of Technology

B.Tech in Computer Science and Engineering

Aug 2023 – Jul 2027

CGPA: 9.36

Coursework: Data Structures & Algorithms, Machine Learning, Artificial Intelligence, Probability & Statistics, Database Systems

TECHNICAL SKILLS

Languages: C++, Python, Java, SQL

AI/ML: PyTorch, TensorFlow, Scikit-learn, XGBoost, SHAP

Generative AI: LLMs, Agentic AI, RAG, LangChain, LangGraph, LangSmith, MCP, Hugging Face

Computer Vision: OpenCV, YOLOv8, OCR

Systems: FastAPI, PostgreSQL, MongoDB, Redis, FAISS, ChromaDB, AWS, Docker, Git, Linux

EXPERIENCE

Physics Wallah – AI Engineer Intern (Remote) [Internship Letter](#)

Mar 2025 – Aug 2025

- Designed and developed **NutriBaba**, an Agentic AI-powered nutrition platform integrating **Multimodal LLMs, RAG, MCP, and LangGraph** for autonomous meal understanding and personalized nutritional reasoning.
- Engineered a scalable AI backend using **FastAPI, PostgreSQL, Redis, ChromaDB, and AWS**, integrating **5+ domain-specific MCP servers** for nutrition, recipes, grocery, fitness, wearable, and user-profile services.
- Built a multi-agent orchestration pipeline with specialized Nutrition, Health Reasoning, Memory, and Explainability Agents, enabling personalized nutritional recommendations and evidence-grounded conversational assistance.

PROJECTS

MediScope AI — Multimodal Clinical Intelligence Platform [GitHub](#)

Spring 2026

- Designed a **dual-input deep learning architecture** in **PyTorch** combining structured clinical features and unstructured medical reports for explainable **multi-label disease prediction** using **SHAP**.
- Built an end-to-end **multimodal AI pipeline** integrating **OCR, Clinical NLP, FAISS Vector Search, Retrieval-Augmented Generation (RAG), and Explainable AI**, enabling semantic retrieval and automated clinical question answering over medical reports.

ShakespeareGPT – Decoder-only Transformer from Scratch [GitHub](#)

Fall 2025

- Implemented a **decoder-only GPT Transformer** from scratch in TensorFlow based on the **Google Brain co-authored “Attention Is All You Need”** paper, implementing **Multi-Head Self-Attention, Positional Encoding, Layer-Norm, Residual Connections, Feed Forward Networks, and Causal Masking**.
- Built a configurable training and inference pipeline with **TensorBoard monitoring, perplexity evaluation, attention visualization, Top-k/Top-p decoding, and autoregressive text generation**, without relying on high-level Transformer APIs.

SATzillaX – Conflict-Driven Clause Learning SAT Solver [GitHub](#)

Fall 2024

- Developed a high-performance **CDCL SAT Solver** in **C++17**, implementing **VSIDS, 1-UIP Conflict Analysis, Two-Watched Literals, Phase Saving, Geometric Restarts, and Non-Chronological Backtracking**.
- Optimized Boolean Constraint Propagation and clause learning for efficient solving of **DIMACS CNF** benchmarks through advanced heuristic optimization and search-space reduction.

ACHIEVEMENTS

- Solved **1150+ algorithmic problems** across LeetCode, Codeforces, GeeksforGeeks, HackerRank, and AtCoder; ranked among the **Top 7.51% globally on LeetCode** (800+ solved) and achieved a **Codeforces Rating of 1389**.
- Secured **2nd Place** in the **Bolt 2.0 Hackathon (Freshers Category, 2024)**.

CERTIFICATIONS

Machine Learning Specialization (Andrew Ng)

IBM Advanced Generative AI Professional Certificate

LEADERSHIP & ACTIVITIES

- Mentored **50+ participants** in astrophotography during **Cosmic Walk 3.0** as part of VIT Stellar.
- Selected for an exclusive **1-on-1 Lightning Session** with **Praveen Jayakumar**, Technology Head – Generative & Agentic AI, AWS, at **AWS Summit Mumbai 2026** on building production-ready Agentic AI systems.